

module3_lab2

September 20, 2026

```
[1]: import pandas as pd
```

```
[2]: df=pd.read_csv("Downloads/usarrests.csv")
```

```
[3]: df.head()
```

```
[3]:   Unnamed: 0  Murder  Assault  UrbanPop
0     Alabama    13.2    236.0        58
1      Alaska    10.0    263.0        48
2     Arizona     8.1    294.0        80
3   Arkansas     8.8    190.0        50
4  California     9.0    276.0        91
```

```
[4]: df.shape
```

```
[4]: (50, 4)
```

```
[5]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Unnamed: 0      50 non-null    str
1   Murder          50 non-null    float64
2   Assault         49 non-null    float64
3   UrbanPop        50 non-null    int64
dtypes: float64(2), int64(1), str(1)
memory usage: 2.1 KB
```

```
[6]: df.describe
```

```
[6]: <bound method NDFrame.describe of      Unnamed: 0  Murder  Assault  UrbanPop
0     Alabama    13.2    236.0        58
1      Alaska    10.0    263.0        48
2     Arizona     8.1    294.0        80
3   Arkansas     8.8    190.0        50
4   California     9.0    276.0        91
```

5	Colorado	7.9	204.0	78
6	Connecticut	3.3	110.0	77
7	Delaware	5.9	238.0	72
8	Florida	15.4	335.0	80
9	Georgia	17.4	NaN	60
10	Hawaii	5.3	46.0	83
11	Idaho	2.6	120.0	54
12	Illinois	10.4	249.0	83
13	Indiana	7.2	113.0	65
14	Iowa	2.2	56.0	570
15	Kansas	6.0	115.0	66
16	Kentucky	9.7	109.0	52
17	Louisiana	15.4	249.0	66
18	Maine	2.1	83.0	51
19	Maryland	11.3	300.0	67
20	Massachusetts	4.4	149.0	85
21	Michigan	12.1	255.0	74
22	Minnesota	2.7	72.0	66
23	Mississippi	16.1	259.0	44
24	Missouri	9.0	178.0	70
25	Montana	6.0	109.0	53
26	Nebraska	4.3	102.0	62
27	Nevada	12.2	252.0	81
28	New Hampshire	2.1	57.0	56
29	New Jersey	7.4	159.0	89
30	New Mexico	11.4	285.0	70
31	New York	11.1	254.0	6
32	North Carolina	13.0	337.0	45
33	North Dakota	0.8	45.0	44
34	Ohio	7.3	120.0	75
35	Oklahoma	6.6	151.0	68
36	Oregon	4.9	159.0	67
37	Pennsylvania	6.3	106.0	72
38	Rhode Island	3.4	174.0	87
39	South Carolina	14.4	879.0	48
40	South Dakota	3.8	86.0	45
41	Tennessee	13.2	188.0	59
42	Texas	12.7	201.0	80
43	Utah	3.2	120.0	80
44	Vermont	2.2	48.0	32
45	Virginia	8.5	156.0	63
46	Washington	4.0	145.0	73
47	West Virginia	5.7	81.0	39
48	Wisconsin	2.6	53.0	66
49	Wyoming	6.8	161.0	60>

```
[7]: df.describe()
```

```
[7]:
```

	Murder	Assault	UrbanPop
count	50.00000	49.000000	50.00000
mean	7.78800	182.183673	74.20000
std	4.35551	130.877435	73.40828
min	0.80000	45.000000	6.00000
25%	4.07500	109.000000	53.25000
50%	7.25000	159.000000	66.00000
75%	11.25000	249.000000	77.75000
max	17.40000	879.000000	570.00000

```
[8]: #Urban population maximum precentage is impossible at 570% as a max value
      ↪posing as an outlier to the data - will need to move the decimal place 1
      ↪place over to 57%
      #Minimum murder per 100,000 is less than 1 at .8, can not have partial murder -
      ↪will change value to 1
      #879 assult is an outlier that fall way outside the 75% range of 249 - can't
      ↪say it is impossible but value is up for review
```

```
[9]: df.isnull().sum()
```

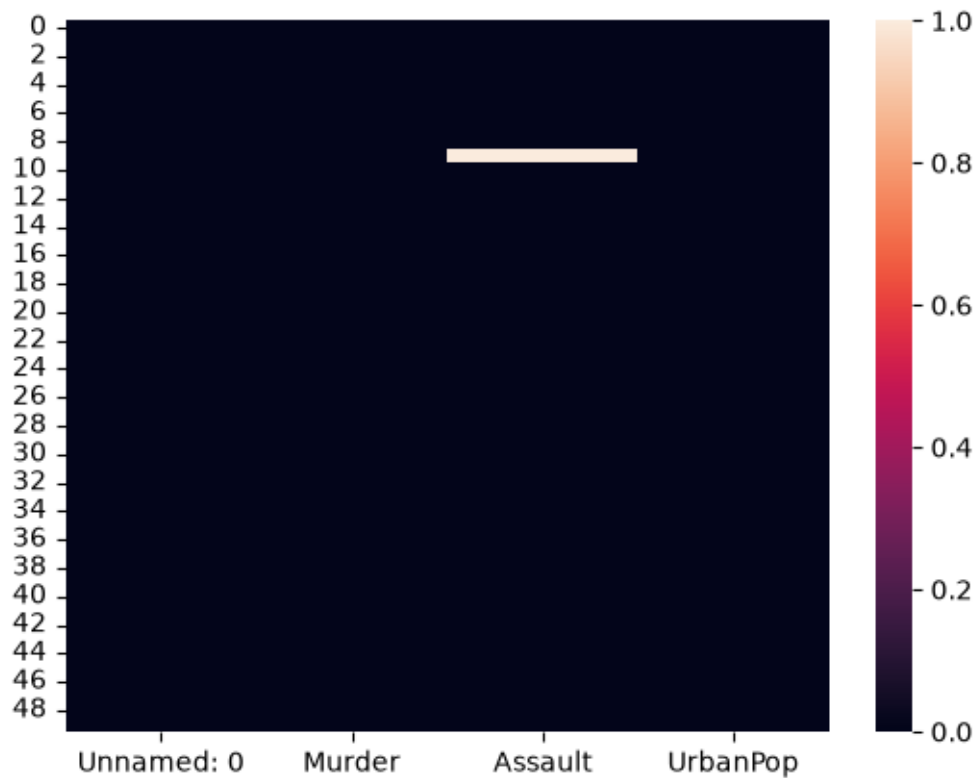
```
[9]: Unnamed: 0      0
      Murder        0
      Assault       1
      UrbanPop      0
      dtype: int64
```

```
[10]: #There is a missing value for assult in Georgia that will need to be filled in.
      ↪Will most likely leave blank til more infomration is gathered
```

```
[11]: import seaborn as sns
```

```
[12]: sns.heatmap(df.isnull())
```

```
[12]: <Axes: >
```



```
[13]: print(df[df['UrbanPop']==570])
```

```
   Unnamed: 0  Murder  Assault  UrbanPop
14         Iowa     2.2    56.0      570
```

```
[14]: #narrow down the outlier of 570 urban pop that would need to be under review
      ↪but can be changed to 57 for possible decimal error
```

```
[15]: df.loc[14, 'UrbanPop']=57.0
```

```
[16]: print(df.loc[14])
```

```
Unnamed: 0    Iowa
Murder        2.2
Assault       56.0
UrbanPop      57
Name: 14, dtype: object
```

```
[17]: df.describe()
```

```
[17]:
```

	Murder	Assault	UrbanPop
count	50.00000	49.000000	50.000000
mean	7.78800	182.183673	63.940000

std	4.35551	130.877435	16.453286
min	0.80000	45.000000	6.000000
25%	4.07500	109.000000	53.250000
50%	7.25000	159.000000	66.000000
75%	11.25000	249.000000	76.500000
max	17.40000	879.000000	91.000000

```
[18]: #change of urbanpop outlier to 57 for Iowa as there is no 570%
```

```
[19]: print(df[df['UrbanPop']==6])
```

	Unnamed: 0	Murder	Assault	UrbanPop
31	New York	11.1	254.0	6

```
[20]: #suspect value of New York being so low considering it being a very populated
      ↪city and state, would flag for review
```

```
[21]: high_assault=df[df['Assault']>249]
```

```
[22]: print(high_assault)
```

	Unnamed: 0	Murder	Assault	UrbanPop
1	Alaska	10.0	263.0	48
2	Arizona	8.1	294.0	80
4	California	9.0	276.0	91
8	Florida	15.4	335.0	80
19	Maryland	11.3	300.0	67
21	Michigan	12.1	255.0	74
23	Mississippi	16.1	259.0	44
27	Nevada	12.2	252.0	81
30	New Mexico	11.4	285.0	70
31	New York	11.1	254.0	6
32	North Carolina	13.0	337.0	45
39	South Carolina	14.4	879.0	48

```
[23]: #South caronlina would be suspcious as it falls outside of the std range and
      ↪would need investigation
```

```
[ ]: #Clean log
      #Changed value to Iowa 570 oercent to 57 for possible decimal error
      #leaving missing value for Georgia until more data is present
      #New York population percentage lower than expected, will need to verify at
      ↪later time if typo as it is an extemere outlier
```